ANALYTICAL REPORT

Diversification Across Six Major Crypto Assets

How much volatility an equal-weight holding actually removes, measured on a 113-day complete-day panel

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Repository	crypto-sql-warehouse
Panel	113 complete trading days, 2026-04-10 to 2026-08-02, 6 assets
Source	`marts.daily_metrics` on PostgreSQL 17.10
Source of figures	`reports/metrics.json`, regenerated by `python scripts/run_analysis.py`

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1. Question and why it matters

Holding several assets instead of one reduces risk only to the extent that those assets move independently. This report measures how much of that reduction six major crypto assets actually deliver. The assets are ADA, BTC, DOGE, ETH, LINK and SOL, all USD spot pairs from a single exchange, held in equal weight at 16.67% each.

The reason to measure it rather than assume it is that the usual case for holding several coins rests on them being different assets with different drivers. If their daily returns move together, an equal-weight basket is close to a single position spread across 6 tickers, and the extra fees and operational surface of 6 holdings buy very little risk reduction.

On this panel the equal-weight portfolio carries an annualised volatility of 0.4579, against 0.5021 for the average single asset. That is 8.8% of the volatility removed. Had the same 6 assets carried the same variances with zero correlation between them, the portfolio would have come out at 0.2087, removing 58.4%. The portfolio captures 15.1% of the reduction that independence between the same 6 variances would have given.

Two caveats belong with that number rather than after it. The panel is 113 days, short enough that the ordering of individual pairs in the correlation matrix is not established by this sample. And the window ends below where it started for most of these assets, so the 8.8% is measured partly through a joint selloff, which is when correlations run highest. Section 5 gives both in full.

> Every correlation in this report is computed on one basis: the 113-day panel of days where all 6 assets have a clean daily return. Section 2 states how that panel is built and section 5 states what it cannot support.

2. Data and method

2.1 Source and grain

Every number comes from `marts.daily_metrics` in the warehouse this repository builds. That mart holds one row per asset per UTC trading day, rolled up in SQL from 16,878 hourly candles in `core.ohlcv_hourly` into 708 daily bars in `marts.daily_ohlcv`. Daily bars are derived from hourly ones rather than fetched separately, so any disagreement between the two granularities can only be a rollup bug, which is findable, and never a source discrepancy, which is not.

The day boundary is pinned to UTC in the rollup expression rather than left to the session timezone. A plain cast would rebucket the whole mart for a client connecting from another zone, which is a bug this warehouse has already had once and now has a regression test for.

2.2 The completeness rule, and why partial days are excluded

The loaded window starts and ends part-way through a day, and the exchange has real outages, so some asset-days are built from fewer than 24 hourly candles. 18 of the 708 daily rows are partial in that way.

A partial day is a different measurement, not a smaller one. Its close is whatever the last observed hour printed rather than the 23:00 candle, and its volume is the sum of however many hours happened to land. Whether it reads as an up day or a down day depends on when the loader ran rather than on the market. Any variance or correlation computed over such a bar inherits that distortion. The mart therefore reports the observed OHLCV on those days and sets every derived column, including `log_return`, to NULL. It does the same on any day that is not adjacent to the previous observation, because a return computed across a gap covers more than one day while carrying a daily label.

2.3 Building one panel

A covariance matrix has to describe one portfolio over one set of dates, so the panel keeps only dates where all 6 assets carry a non-null log return. Of 118 observed dates, 5 are excluded: 3 carry at least one partial bar (2026-04-08, 2026-05-08 and 2026-08-03), which disqualifies the date for all 6 assets at once, and 2 are the first observation after one of those, where the adjacency guard applies. Each exclusion removes the date for all

6 assets at once, so the panel is not lopsided toward any name. What remains is 113 days from 2026-04-10 to 2026-08-02, 678 asset-days with no holes.

The alternative, computing each correlation over whatever dates that particular pair happened to share, lets different entries of the matrix describe different periods. A matrix assembled that way is not guaranteed to be positive semidefinite, and the portfolio variance it produces can come out negative. Dropping 5 dates for everyone is the cheaper problem.

2.4 Returns, annualisation and covariance

Returns are logarithmic, ln(close today / close yesterday), because they add across periods and are the standard input to a variance estimate. Volatility is the sample standard deviation of those daily returns with the n minus 1 divisor, multiplied by the square root of 365.

The 365 is deliberate. Crypto trades every calendar day including weekends and holidays, so a year holds 365 return observations. The square root of 252 that equity work uses counts exchange sessions, and a market with no sessions off does not have 252 of them. Annualising this data on 252 would report every volatility in this report about 17% lower than it is.

The covariance matrix is the sample covariance of the same daily log returns, also with the n minus 1 divisor, scaled by 365. Portfolio volatility is the square root of w'Cw with w set to 0.1667 in every position. The zero-correlation counterfactual replaces C with its own diagonal: every asset keeps its variance exactly, and every covariance is set to zero. Independence is a benchmark, not a floor. Negative covariances put a portfolio below it, which is what a hedge is for, and `tests/test_analysis.py` checks that the arithmetic behaves that way. Six long spot positions in the same sector will not supply negative covariance, so for the case at hand the independent portfolio is the reference the real number is measured against.

2.5 One basis, stated once

The warehouse also publishes `marts.pairwise_correlations_90d`, which reports 0.739 to 0.928 across its 15 pairs. That mart runs on a trailing 90-day window and lands 87 to 87 shared days per pair, so it is a different window on the same data and its numbers are not interchangeable with the ones below. Nothing in this report uses them.

3. Results

3.1 How closely the assets move together

All 15 pairs sit between 0.726 and 0.895, averaging 0.800. The loosest pair is ADA and BTC at 0.726 and the tightest is ETH and LINK at 0.895. There is no pair in this set that a portfolio could lean on for independence.

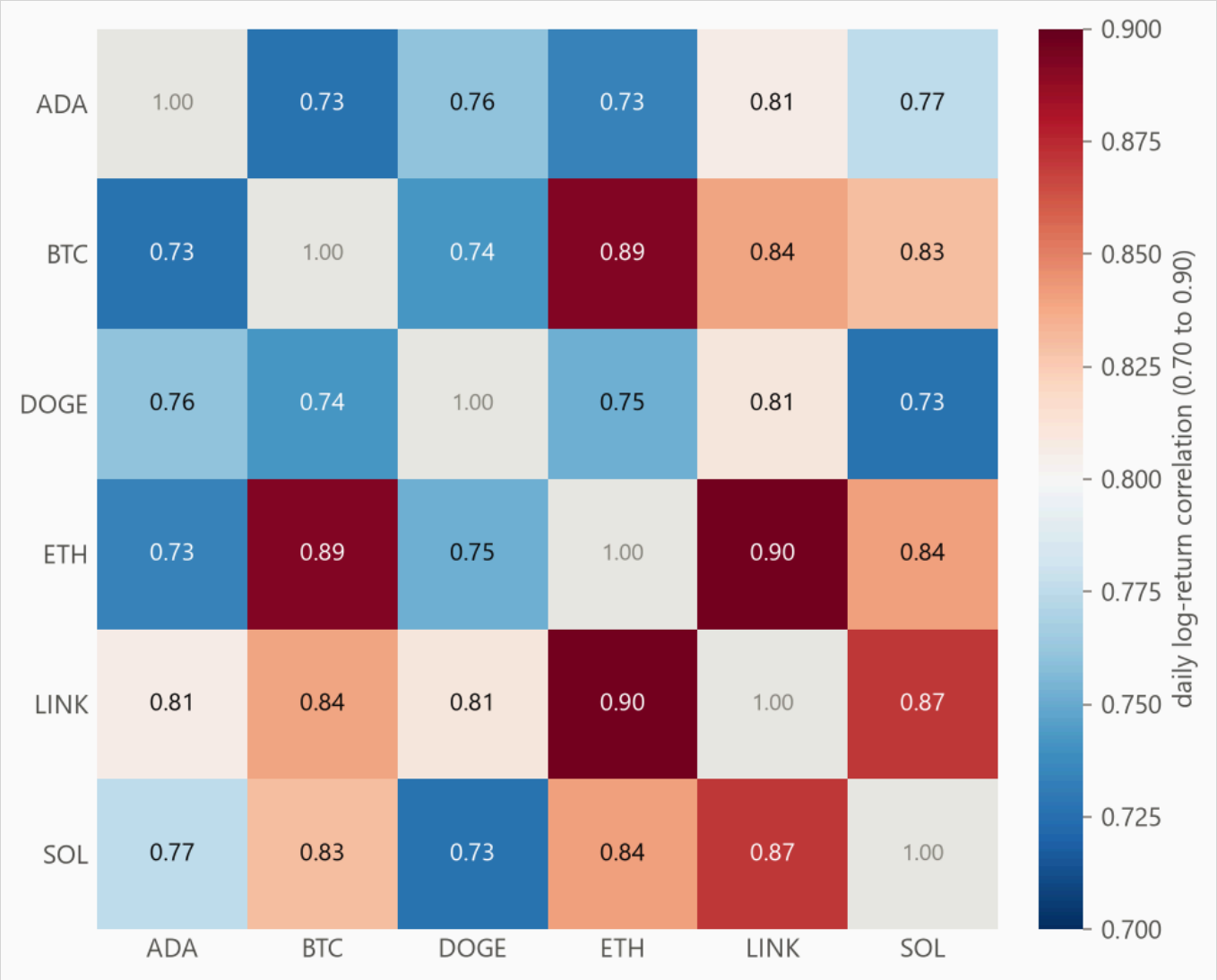


Figure 1. Pairwise correlation of daily log returns over the 113-day panel. The colour axis runs from 0.70 to 0.90 rather than the full -1 to 1 range, because on a full scale every cell here would render in nearly the same tone; read the colours as differences within a tight band, not as a general correlation scale. The diagonal is blanked, since a variable's correlation with itself is 1 by construction. The assets form one block: the visible variation runs from 0.726 to 0.895, which is a band inside the high end rather than a range from low to high.

Table 1. The full correlation matrix behind Figure 1, on the 113-day complete-day panel.

	ADA	BTC	DOGE	ETH	LINK	SOL
ADA	1.00	0.726	0.762	0.735	0.805	0.775
BTC	0.726	1.00	0.742	0.891	0.839	0.830
DOGE	0.762	0.742	1.00	0.752	0.810	0.727
ETH	0.735	0.891	0.752	1.00	0.895	0.841
LINK	0.805	0.839	0.810	0.895	1.00	0.870
SOL	0.775	0.830	0.727	0.841	0.870	1.00

3.2 How volatile the assets are individually

Annualised volatility ranges from 0.357 for BTC to 0.684 for ADA, averaging 0.502. That average is the reference the portfolio has to beat, because it is what an investor holding one of these names at random would have carried.

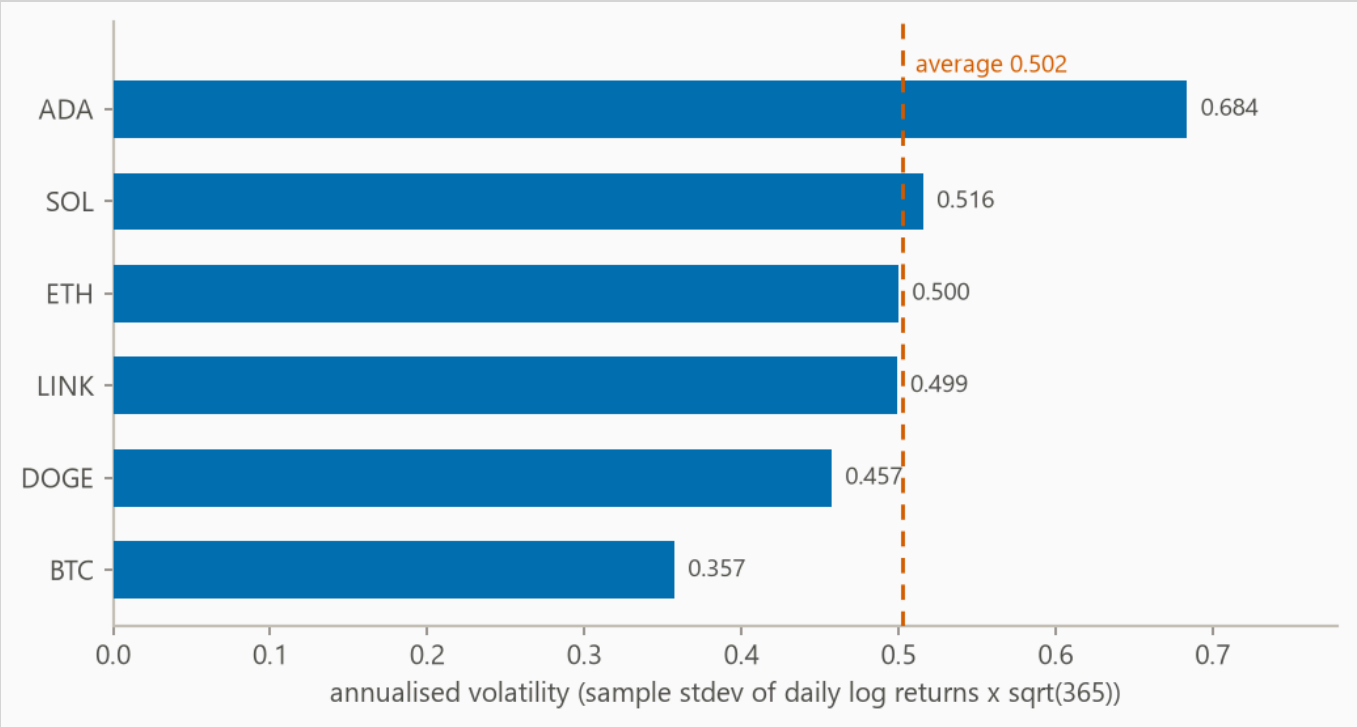


Figure 2. Annualised volatility per asset, sorted, with the 0.502 average marked. What to conclude: the assets are not interchangeable in risk terms. ADA carries roughly 1.9 times the volatility of BTC, so equal weight is not the same thing as equal risk contribution.

Table 2. Per-asset figures over the panel. Drawdown is measured from the running peak within the loaded window, not from an all-time high. Total log return is the sum of the panel's daily log returns.

ASSET	ANNUALISED VOLATILITY	MAX DRAWDOWN	TOTAL LOG RETURN
ADA	0.684	-49.2%	-0.325
SOL	0.516	-36.2%	-0.177
ETH	0.500	-36.3%	-0.166
LINK	0.499	-32.9%	-0.118
DOGE	0.457	-40.3%	-0.277
BTC	0.357	-28.8%	-0.131

3.3 What the co-movement looks like over time

The correlation matrix states the co-movement as 15 numbers. The drawdown paths show the same fact as a picture: the assets fall together and recover together, and the differences between them are mostly depth rather than timing.

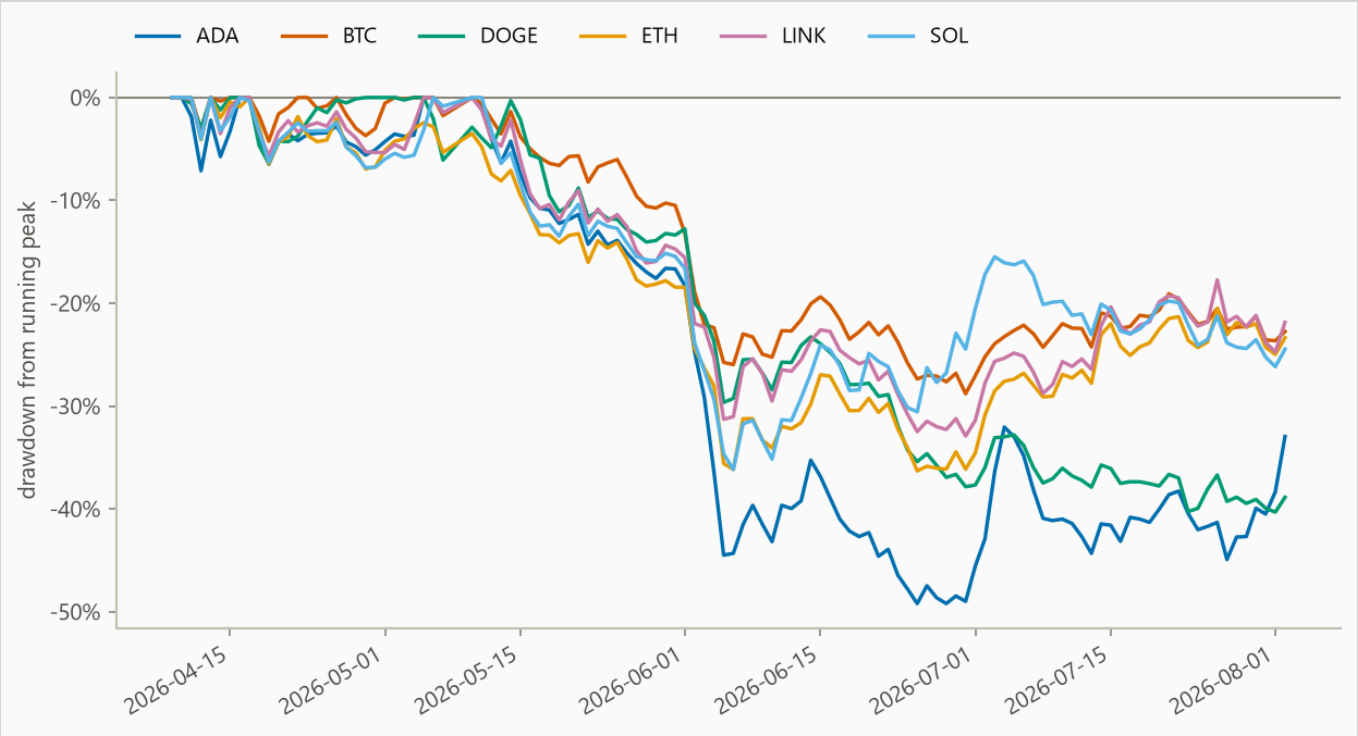


Figure 3. Each asset's drawdown from its own running peak, all on one axis. What to conclude: the troughs line up in time. The deepest reading is -49.2% for ADA, and no asset in this set was rising while the others fell, which is what would have to happen for the basket to be meaningfully safer than its members.

3.4 The diversification result

Equal weights on the 113-day covariance matrix give an annualised portfolio volatility of 0.4579. The average single asset carried 0.5021, so the basket removed 8.8% of the volatility. Holding the same 6 variances with every correlation set to zero would have given 0.2087, a reduction of 58.4%. The ratio of the two, 15.1%, is the share of the available diversification these assets actually delivered.

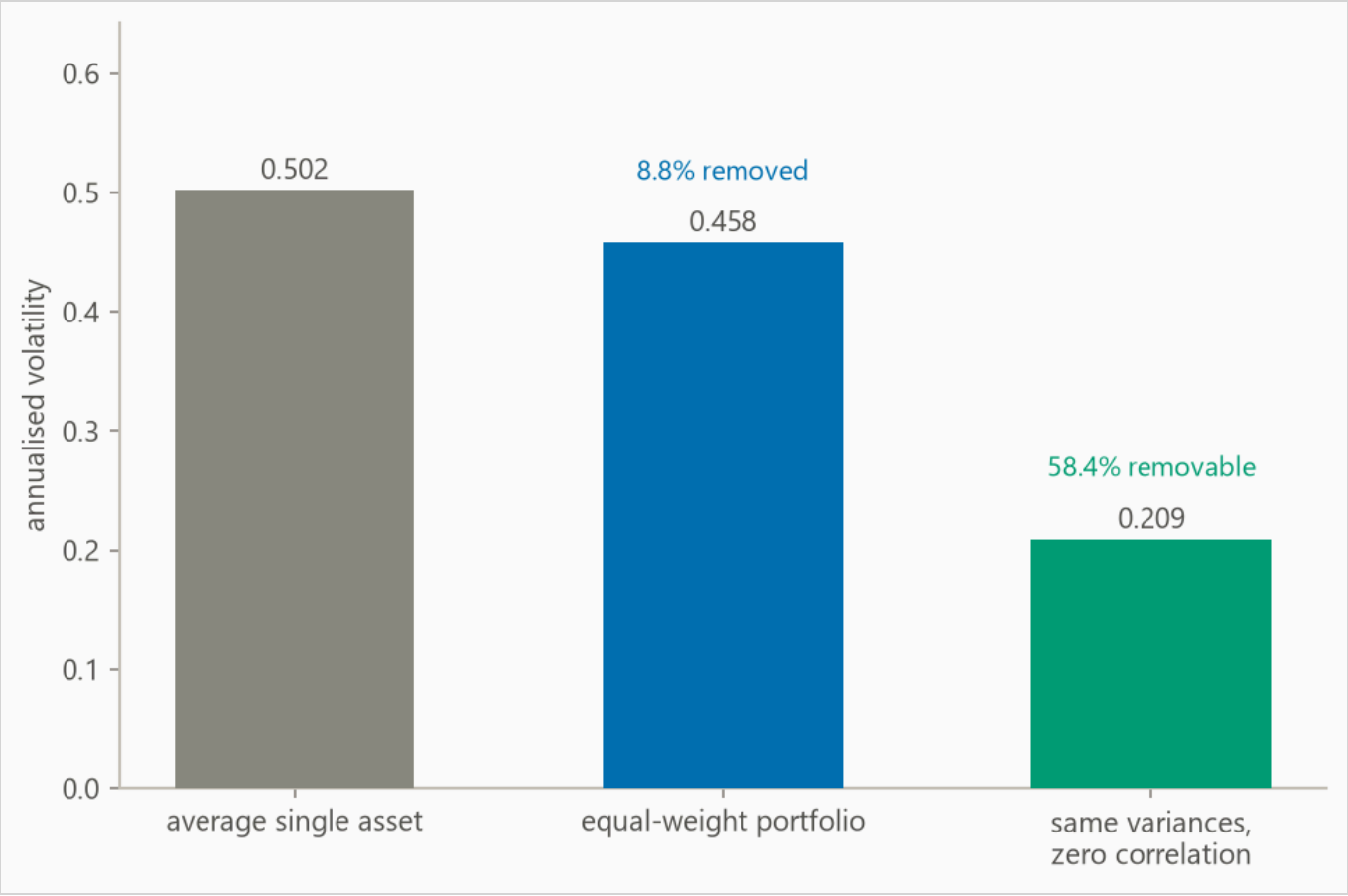


Figure 4. Three annualised volatilities: on the left the average of holding one of these assets, in the middle the equal-weight portfolio, on the right the same 6 variances with zero correlation. The left-to-middle gap is the benefit received and the middle-to-right gap is what the correlations consumed. Read the position of the middle bar: it sits close to the left one and far from the right one.

Table 3. The four volatility figures and the ratios derived from them, all annualised.

QUANTITY	VALUE
Average single-asset volatility	0.5021
Equal-weight portfolio volatility	**0.4579**
Same variances, zero correlation	0.2087
Risk removed by the portfolio	8.8%
Risk removable at zero correlation	58.4%
Share of available diversification captured	**15.1%**

4. What this means in practice

Spreading a holding across these 6 names removed 8.8% of its volatility. Counting names is therefore close to useless as a risk control here. Independence between the same 6 variances would have removed 58.4%, so the shortfall sits in the correlations and not in the number or the size of the positions. The correlations that hold deliver 15.1% of it. An equal-weight basket of equally correlated assets converges toward the average volatility times the square root of the mean correlation, which is 0.449 here against today's 0.458, so a seventh name from the same group would buy very little.

The sharper version of the same point is in Table 2. BTC alone carried 0.357, which is below the equal-weight basket's 0.458. Over this window, spreading into the other 5 names raised volatility rather than lowering it, because the diversification benefit was smaller than the extra variance those names brought with them. Equal weight assumes the assets are interchangeable, and on this evidence they are not.

Two consequences follow for anyone using this warehouse. First, position size and hedging do the work that name count does not: at a mean correlation of 0.800, total exposure is the variable that moves portfolio risk. Second, the quantity worth monitoring is the correlation level itself, which is what `marts.pairwise_correlations_90d` exists to track on a rolling basis. Only a fall in that correlation would change the conclusion here.

None of this is an argument about which assets to hold. It measures what the 6 names already loaded did over one window. Section 5 states what that window does not cover.

5. Limitations

The panel is 113 days. A single pair at the average correlation of 0.800 has a 95% Fisher interval of 0.722 to 0.858 on this many observations, a width of 0.136. The entire spread between the loosest and tightest pair in Figure 1 is 0.170, so the uncertainty on any one correlation is about 80% as wide as the range the heatmap is showing. The ordering of pairs in that figure is therefore not established by this sample. What survives is that the whole block sits high, not that any particular pair sits above any other.

One exchange and one quote currency. All candles come from Coinbase and every pair is quoted in USD. There is no second venue to check the prices against, so a venue-specific outage or a stale feed enters the covariance matrix as if it were market structure. Common exposure to the dollar sits inside every one of these correlations, and with no second quote currency loaded there is no way to separate it out.

Correlations are not stable, and they rise in selloffs. This is the limitation that matters most for how the headline number should be read. Correlation between risk assets rises when they fall together, which is when diversification is being asked to do something. This panel cannot test that on its own, because it holds one joint decline and no calm stretch to compare it against. That decline is visible in Figure 3, and every one of the 6 assets ended the window down. The 8.8% is therefore a reading taken partly through a stressed market, and it is more likely to be optimistic than conservative as a forecast of the next one.

Drawdown is measured from the start of the loaded window. The peak in Table 2 and Figure 3 is the highest cumulative level since 2026-04-10, not an all-time high. Backfilling earlier history would change every drawdown figure in this report. These are not the numbers a trading desk means by maximum drawdown.

Equal weight is not an optimised portfolio. The weights are 0.1667 each because that is the simplest allocation to state, not because anything here selected them. A minimum-variance or risk-parity allocation would produce a different portfolio volatility, and given the 1.9 to 1 spread in single-asset volatility it would probably produce a lower one. The w'Cw calculation also assumes the weights stay at 0.1667, which in practice needs rebalancing, and no transaction costs, funding, borrow or slippage are modelled anywhere in this report.

Six assets is not the market. These are large-cap USD spot pairs, which is one corner of one asset class. I picked them for coverage and liquidity when I built the warehouse. They are not a sample of anything. A result about ADA, BTC, DOGE, ETH, LINK and SOL is a result about those 6 names on this venue over these 113 days, and reading it as a claim about crypto in general would be reading it too widely.

6. How to reproduce

Python 3.11 or later and a PostgreSQL server. This run used PostgreSQL 17.10; nothing in the SQL needs a feature newer than PostgreSQL 10. From the repository root:

```
pip install -r requirements.txt
copy .env.example .env
python scripts/run_setup.py
python scripts/run_ingest.py --backfill-days 365 --refresh-marts
python scripts/run_analysis.py
python scripts/run_build_report.py
```

`run_setup.py` creates the database and applies the schema and the marts. `run_ingest.py` backfills hourly candles and refreshes the marts; the first run pulls the full backfill and every later run resumes from the watermark. `run_analysis.py` builds the panel, writes `reports/metrics.json` and the four figures under `reports/figures/`, and prints the headline numbers to the log. `run_build_report.py` renders this document from that metrics file and prints it to PDF when Chrome is installed. Every measured quantity in the prose above is read from `reports/metrics.json` at render time, so no result here can drift from the analysis that produced it. The fixed conventions the method rests on, the 365-day year and the 24-hour day, are written into the template and stated in section 2.

Three things stand behind the pipeline underneath. `python scripts/run_verify.py` runs 5 checks against a loaded warehouse, each fatal on failure, covering hourly gaps, rollup coverage, raw-to-core agreement, upsert idempotency and run status. `python -m pytest` runs 70 test functions, none of which need a database or a network, covering the ingestion windowing, the API client and the diversification math in this report. The 20 annotated queries in `sql/04_queries.sql` each run standalone against a loaded database.

Rerunning the ingestion loads whatever the exchange has published since, which moves the panel end date and every number here. The figures in this document are the ones produced by the run recorded in `reports/metrics.json`, generated 2026-08-03T21:44:40Z against PostgreSQL 17.10.
